Matrices, Vectors & Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In the standard (x, y) coordinate plane, vector $\mathbf{u} = \langle 2, -5 \rangle$ and vector $\mathbf{v} = \langle -3, 4 \rangle$. What is the vector $2\mathbf{u} + \mathbf{v}$? Show your complete algebraic work.

Final Answer: $\langle 1, -6 \rangle$. To find $2\mathbf{u} + \mathbf{v}$, first perform the scalar multiplication on $\mathbf{u}$: $2\mathbf{u} = 2\langle 2, -5 \rangle = \langle 2(2), 2(-5) \rangle = \langle 4, -10 \rangle$ Next, add the corresponding components of $2\mathbf{u}$ and $\mathbf{v}$: $2\mathbf{u} + \mathbf{v} = \langle 4 + (-3), -10 + 4 \rangle = \langle 1, -6 \rangle$ Common misconceptions: - Computing $\mathbf{u} + \mathbf{v}$ without multiplying by 2 yields.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In the complex plane, the point representing the complex number $z_1 = 3 + 4i$ is labeled A. The complex number z_2 is obtained by rotating z_1 counterclockwise by 90° about the origin, and its corresponding point is labeled B. The complex number z_3 is the complex conjugate of z_1, and its corresponding point is labeled C. What is the area of ΔABC? Show your complete algebraic work.

Final Answer: 28. 1. **Find point A:** The complex number $z_1 = 3 + 4i$ corresponds to the point $A(3, 4)$ in the complex plane. 2. **Find point B:** Rotating a complex number counterclockwise by 90° about the origin is equivalent to multiplying it by i: $z_2 = i(3 + 4i) = 3i + 4i^2 = -4 + 3i$ Thus, point B has coordinates $(-4, 3)$. 3. **Find point C:** The complex conjugate of $z_1 = 3 + 4i$ is $z_3 = \overline{z_1} = 3 - 4i$. Thus, point C has coordinates $(3, -4)$. 4. **Calculate the area of.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the product of $-4i$ and its complex conjugate.

Final Answer: 16. Applying Matrices, Vectors & Complex Numbers principles, $(-4i)(4i) = (0)^2 + (-4)^2 = 16$. Therefore, the verified correct option is 16.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Find the magnitude of the vector $\langle -20, -21 \rangle$.

Final Answer: 29. $|\vec{v}| = \sqrt{(-20)^2 + (-21)^2} = \sqrt{400 + 441} = 29$.

5 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Simplify $(2 + 3i)(1 - i)$ and write in the standard form $a + bi$.

Multiply using FOIL: $(2 + 3i)(1 - i) = 2(1) - 2i + 3i - 3i^2 = 2 + i - 3(-1) = 2 + i + 3 = \mathbf{5 + i}$

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A shirt is marked up by 40% and then placed on sale for 25% off. What is the net percentage increase over the original cost?

Multiply scale factors: $1.40 \times 0.75 = 1.05$, which corresponds to a **5%** net increase.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Does the ACT Math test penalize wrong guesses, or should every single question be answered before time expires?

There is no guessing penalty on the ACT; raw score equals the count of correct answers out of 60. Never leave a bubble blank.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Under the 20-20-20 pacing strategy, how many minutes should remain on the 60-minute clock when you finish Question 20?

You spend 15 minutes on Questions 1–20, leaving $60 - 15 = \mathbf{45}$ minutes remaining.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the ordered dataset of 12 numbers: 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 38, 72. (a) Determine Q_1, the median, Q_3, and the IQR. (b) Use Tukey's $1.5 \times$ IQR rule to determine whether 72 is an outlier.

(a) Five-Number Summary Calculations:Lower half (first 6 values): 14, 16, 18, 20, 22, 24 $\Rightarrow Q_1 = \frac{18+20}{2} = \mathbf{19}$.Median of all 12 values: $\frac{24+26}{2} = \mathbf{25}$.Upper half (last 6 values): 26, 28, 30, 34, 38, 72 $\Rightarrow Q_3 = \frac{30+34}{2} = \mathbf{32}$.IQR $= Q_3 - Q_1 = 32 - 19 = \mathbf{13}$. (b) Outlier Boundary Check:

Upper Outlier Boundary $= Q_3 + 1.5 \times \text{IQR} = 32 + 1.5(13) = 32 + 19.5 = \mathbf{51.5}$. Since $72 > 51.5$, the va.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express the complex number $z = \frac{5+2i}{3-4i}$ in standard form $a + bi$, and determine its modulus $|z|$.

(a) Standard form $a + bi$: Multiply numerator and denominator by the complex conjugate $3 + 4i$:

$z = \frac{(5+2i)(3+4i)}{(3-4i)(3+4i)} = \frac{15+20i+6i+8i^2}{3^2-(4i)^2} = \frac{15+26i-8}{9-(-16)} = \frac{7+26i}{25} = \mathbf{\frac{7}{25} + \frac{26}{25}i}$ (b) Modulus $|z|$: Using the quotient property $|z_1/z_2| = |z_1|/|z_2|$: $|5+2i| = \sqrt{5^2+2^2} = \sqrt{29}$, $\quad |3-4i| = \sqrt{3^2+(-4)^2} = \sqrt{25} = 5$ $|z| = \frac{\sqrt{29}}{5} \approx 1.077$

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11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

In a course grading system, Homework is weighted 20%, Midterm Exam is weighted 35%, and Final Exam is weighted 45%. A student scored 92 on Homework and 76 on the Midterm Exam. What is the minimum score the student must achieve on the Final Exam to earn an overall course average of at least 84?

Step 1: Set up the weighted average inequality with final exam score F: $0.20(92) + 0.35(76) + 0.45(F) \geq 84$. Step 2: Compute the accumulated points: $0.20(92) = 18.4, \quad 0.35(76) = 26.6 \Rightarrow 18.4 + 26.6 = 45.0$. Step 3: Isolate F: $45.0 + 0.45F \geq 84 \Rightarrow 0.45F \geq 84 - 45 = 39 \Rightarrow F \geq \frac{39}{0.45} = \frac{3900}{45} = \frac{260}{3} \approx 86.67$. The student must score at least $\mathbf{86.7}$ (or $\mathbf{\frac{260}{3}}$) on the Final Exam.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Let matrices $A = (3, -1; 2, 4)$ and $B = (1, 5; 0, -2)$. Compute AB, $\det(A)$, $\det(B)$, and verify that $\det(AB) = \det(A)\det(B)$.

(a) Product AB: $AB = ((3)(1) + (-1)(0), (3)(5) + (-1)(-2); (2)(1) + (4)(0), (2)(5) + (4)(-2)) = (\mathbf{3, 17; 2, 2})$ (b) Determinants: $\det(A) = (3)(4) - (-1)(2) = 12 + 2 = 14$. $\det(B) = (1)(-2) - (5)(0) = -2$. Product of determinants: $14 \times (-2) = -28$. From AB: $\det(AB) = (3)(2) - (17)(2) = 6 - 34 = -28$. Thus $\det(AB) = \det(A)\det(B) = -28$. ✓

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Evaluate the complex expression $\frac{(3-2i)^2}{1+i}$ and write the result in standard form $a + bi$.

Step 1: Expand the numerator: $(3-2i)^2 = 3^2 - 2(3)(2i) + (2i)^2 = 9 - 12i + 4(-1) = 5 - 12i$. Step 2: Multiply by conjugate $1 - i$: $\frac{5-12i}{1+i} = \frac{(5-12i)(1-i)}{(1+i)(1-i)} = \frac{5-5i-12i+12i^2}{1-i^2} = \frac{5-17i-12}{1-(-1)} = \frac{-7-17i}{2}$. In standard form: $\mathbf{-\frac{7}{2} - \frac{17}{2}i}$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Section I of EST I Math contains 20 questions in 25 minutes. If a student answers the first 12 questions in an average of 50 seconds each, how many seconds per question are available for the remaining 8 questions?

Time for first 12 questions $= 12 \times 50 = 600$ seconds (10 minutes). Total time $= 25 \times 60 = 1500$ seconds. Remaining time $= 1500 - 600 = 900$ seconds. For the remaining 8 questions: $\frac{900}{8} = 112.5$ seconds per question (1 minute 52.5 seconds per question).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

The ACT Subject Test in Mathematics consists of 50 questions in 60 minutes. If a student spends an average of 55 seconds on each of the first 30 questions, how many seconds per question are available for the remaining 20 questions?

Time spent on first 30 questions = $30 \times 55 = 1650$ seconds (27.5 minutes).
Total time available = $60 \times 60 = 3600$ seconds. Remaining time = $3600 - 1650 = 1950$ seconds. For the final 20 questions: $\frac{1950}{20} = 97.5$ seconds per question.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Level 1 · FoundationsThe Digital SAT Math section allocates 70 minutes across two modules of equal length. If each module contains 22 questions, what is the exact average time available per question in seconds?

Step 1: Calculate total seconds per module: $35 \text{ minutes} \times 60 \text{ s/min} = 2100 \text{ seconds}$. Step 2: Divide by the 22 questions in the module: $\frac{2100 \text{ seconds}}{22 \text{ questions}} = \frac{1050}{11} \approx 95.45$ seconds (about 1 min 35 s per question).

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Functions & Graphs

Level 4 · StrategyA student finishes the first 18 questions of Module 1 with 9 minutes remaining. Questions 19 and 20 are lengthy geometry proofs, while Questions 21 and 22 are quick Desmos system intersection problems. In what order should the student attack the remaining 4 questions, and why?

Every Digital SAT question carries equal raw weight within a module. The student should immediately flag Questions 19–20, solve Questions 21–22 in under 90 seconds using Desmos, and spend the remaining 7.5 minutes on Questions 19–20 without time pressure.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

Level 2 · MediumA student solves a linear equation and obtains the fraction $x = -\frac{7}{4}$. What are the valid ways to enter this answer into the Digital SAT Student-Produced Response (SPR) grid?

The Digital SAT SPR allows up to 5 characters including negative signs and decimal points:As a fraction: -7/4 (4 characters).As a terminating decimal: -1.75 (5 characters).Invalid: Mixed number -1 3/4 is read as -13/4 and will be graded incorrect.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Plane Geometry

Given vectors $\vec{u} = \langle 3, -4, 2 \rangle$ and $\vec{v} = \langle 2, 1, k \rangle$, find the value of k such that $\vec{u}$ and $\vec{v}$ are orthogonal.

Two non-zero vectors are orthogonal if and only if their dot product equals zero: $\vec{u} \cdot \vec{v} = 0$
$(3)(2) + (-4)(1) + (2)(k) = 0$ $6 - 4 + 2k = 0 \Rightarrow 2 + 2k = 0 \Rightarrow 2k = -2 \Rightarrow \mathbf{k = -1}$

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Trigonometry

The mean of 10 numbers is 15. One number is removed and the new mean of the remaining 9 numbers is 14. Find the removed number.

Step 1: Total sum of the original 10 numbers: $S_{10} = 10 \times 15 = 150$ Step 2: Total sum of the remaining 9 numbers: $S_9 = 9 \times 14 = 126$ Step 3: The removed value is: $x = S_{10} - S_9 = 150 - 126 = \mathbf{24}$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the product of the complex numbers $(4 - 3i)$ and $(2 + 5i)$, where $i^2 = -1$?

a. $23 + 14i$ ✓

b. $7 + 14i$

c. $23 - 14i$

d. $8 - 15i$

Expand: $4(2) + 4(5i) - 3i(2) - 15i^2 = 8 + 20i - 6i - 15(-1) = 8 + 14i + 15 = 23 + 14i$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the simplified form of the quotient $\frac{5 - i}{2 + i}$?

a. $\frac{9 - 7i}{5}$ ✓

b. $\frac{11 - 7i}{5}$

c. $3 - i$

d. $\frac{9 + 3i}{5}$

Multiply numerator and denominator by conjugate $(2 - i)$: $\frac{(5 - i)(2 - i)}{(2 + i)(2 - i)} = \frac{10 - 5i - 2i + i^2}{4 - i^2} = \frac{10 - 7i - 1}{4 - (-1)} = \frac{9 - 7i}{5}$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the value of i^{47}, where $i = \sqrt{-1}$?

a. 1

b. -1

c. i

d. $-i$ ✓

Divide power by 4: $47 = 4(11) + 3$. Thus $i^{47} = (i^4)^{11} \cdot i^3 = 1 \cdot (-i) = -i$.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the modulus (absolute value) of the complex number $z = -5 + 12i$?

a. 7

b. 13 ✓

c. 17

d. $\sqrt{119}$

Modulus $|z| = \sqrt{a^2 + b^2} = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the distance in the complex plane between $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$?

a. $2\sqrt{13}$ ✓

b. $\sqrt{20}$

c. 10

d. 5

Distance $= |z_1 - z_2| = |(3 - (-1)) + (-2 - 4)i| = |4 - 6i| = \sqrt{4^2 + (-6)^2} = \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $z = 2 + 2i$, what is the argument θ of z in radians, where $0 \le \theta < 2\pi$?

a. $\frac{\pi}{6}$

b. $\frac{\pi}{4}$ ✓

c. $\frac{\pi}{3}$

d. $\frac{3\pi}{4}$

Both real and imaginary parts are positive (Quadrant I). $\tan\theta = \frac{b}{a} = \frac{2}{2} = 1 \Rightarrow \theta = \frac{\pi}{4}$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the determinant of the 2×2 matrix $M = [[4, 6], [-2, 3]]$?

a. 0

b. 12

c. 24 ✓

d. 36

Determinant $\det(M) = ad - bc = 4(3) - (6)(-2) = 12 - (-12) = 24$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of k will the matrix $K = [[k, 6], [2, 3]]$ NOT have an inverse?

a. 1

b. 2

c. 3

d. 4 ✓

A matrix is singular (no inverse) when its determinant is zero:

$k(3) - (6)(2) = 0 \Rightarrow 3k - 12 = 0 \Rightarrow 3k = 12 \Rightarrow k = 4$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$ are given. What are the components of $2\mathbf{u} + \mathbf{v}$?

a. $\langle 4, -3 \rangle$ ✓

b. $\langle 1, 1 \rangle$

c. $\langle 4, 3 \rangle$

d. $\langle -1, -3 \rangle$

$2\mathbf{u} = \langle 6, -8 \rangle$. Adding $\mathbf{v}$: $\langle 6 + (-2), -8 + 5 \rangle = \langle 4, -3 \rangle$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the dot product of the vectors $\mathbf{a} = \langle 5, -2 \rangle$ and $\mathbf{b} = \langle 4, 7 \rangle$?

a. 6 ✓

b. 26

c. -6

d. 34

$\mathbf{a} \cdot \mathbf{b} = (5)(4) + (-2)(7) = 20 - 14 = 6$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the magnitude of the vector $\mathbf{w} = \langle -9, 12 \rangle$?

a. 13

b. 15 ✓

c. 17

d. 21

Magnitude $||\mathbf{w}|| = \sqrt{(-9)^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the complex number system, for what real value of k is the expression $\frac{k+8i}{2-5i}$ a purely imaginary number? (Note: $i = \sqrt{-1}$)

a. -20

b. $-\frac{16}{5}$

c. $\frac{16}{5}$

d. 20 ✓

To determine when the fraction is purely imaginary, multiply the numerator and denominator by the complex conjugate of the denominator, $2+5i$: $\frac{k+8i}{2-5i} \cdot \frac{2+5i}{2+5i} = \frac{(k+8i)(2+5i)}{(2)^2-(5i)^2}$ Simplify the denominator: $2^2 - 25i^2 = 4 - 25(-1) = 4 + 25 = 29$ Expand the numerator using FOIL: $(k+8i)(2+5i) = 2k + 5ki + 16i + 40i^2 = 2k + (5k+16)i - 40 = (2k-40) + (5k+16)i$ Thus, the expression can be written in standard form $a+bi$ as.

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the imaginary unit i, where $i^2 = -1$, which of the following complex numbers is equal to $(5-3i)-(2-8i)$?

a. $3+5i$ ✓

b. $3-11i$

c. $7-11i$

d. $3-5i$

To subtract complex numbers, distribute the negative sign to both terms inside the second set of parentheses: $(5-3i)-(2-8i) = 5-3i-2+8i$ Next, combine the real parts and the imaginary parts separately: Real parts: $5-2=3$ Imaginary parts: $(-3+8)i = 5i$ Combining these yields $3+5i$. Common Misconceptions: - $3-11i$ occurs if the negative sign is not distributed to $-8i$, yielding $-3i-8i=-11i$. - $7-11i$ occurs if the real parts are added ($5+2=7$) instead of subtracted. - $3-5i$.

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

If $A = (2, -11; 9, -3)$ and $B = (-14, -11; 11, -5)$, find the entry in row 1, column 2 of $A+B$.

a. -25

b. -21

c. -16

d. **-22** ✓

Applying Matrices, Vectors & Complex Numbers principles, Row 1, Col 2: $-11 + -11 = -22$. Therefore, the verified correct option is -22.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the complex conjugate of $11 - 7i$.

a. 11-7i

b. **11+7i ✓**

c. -11+7i

d. -11-7i

The conjugate negates the imaginary part: $11 + 7i$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the determinant of $(-5, -7; -3, -7)$.

a. 9

b. 18

c. 16

d. **14 ✓**

$\det = (-5)(-7) - (-7)(-3) = 35 - 21 = 14$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Write 1×10^4 in standard (decimal) form.

a. 20000

b. 1000

c. **10000 ✓**

d. 100000

Moving the decimal point 4 places right gives 10000.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

A quantity increases from 15 to 18. What is the percent increase?

a. **20\% ✓**

b. 16\%

c. 29\%

d. 10\%

$\frac{18-15}{15} \times 100 = 20\%$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Plane Geometry

If 6 notebooks cost $x24, what is the cost per notebook?

a. $x23

b. $x33

c. $x1

d. $x**4** ✓

$24 \div 6 = 4$ dollars per notebook.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Trigonometry

A quantity decreases from 10 to 5. What is the percent decrease?

a. 30\%

b. 20\%

c. 24\%

d. **50\%** ✓

$\frac{10-5}{10} \times 100 = 50\%$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Statistics & Probability

If 2 tickets cost $x12, what is the cost per ticket?

a. $x**6** ✓

b. $x7

c. $x8

d. $x17

$12 \div 2 = 6$ dollars per ticket.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

If 3 pencils cost $x9, what is the cost per pencil?

a. $x16

b. $x12

c. $x6

d. $x**3** ✓

$9 \div 3 = 3$ dollars per pencil.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Given $\mathbf{u} = \langle 1, 2, 0 \rangle$ and $\mathbf{v} = \langle 0, 1, 3 \rangle$, what is the cross product $\mathbf{u} \times \mathbf{v}$?

a. $\langle 2, 3, 0 \rangle$

b. $\langle 6, 3, 1 \rangle$

c. $\langle 6, -3, 1 \rangle$ ✓

d. $\langle -6, -3, -1 \rangle$

$\mathbf{u} \times \mathbf{v} = \mathbf{i}(2 \cdot 3) - \mathbf{j}(1 \cdot 3) + \mathbf{k}(1 \cdot 1) = \langle 6, -3, 1 \rangle$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

In the standard (x, y) coordinate plane, a laser beam emitted from point $A(-3, 8)$ reflects off the x-axis at point $P(c, 0)$ and travels directly to target $B(5, 4)$. Given that the angle of incidence equals the angle of reflection, what is the value of c?

a. 1

b. $\frac{5}{3}$

c. $\frac{7}{3}$ ✓

d. $\frac{8}{3}$

We can solve this using either similar right triangles or geometric reflection across the x-axis. **Method 1: Similar Triangles** Drop vertical segments from $A(-3, 8)$ and $B(5, 4)$ to the x-axis. Because the angle of incidence equals the angle of reflection, the angle the incoming ray makes with the x-axis equals the angle the reflected ray makes with the x-axis. Therefore, the two right triangles formed with the x-axis are similar: $\tan\theta = \frac{8}{c-(-3)} = \frac{4}{5-c}$ Cross-multiplying y.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Coordinate Geometry

If $\sin\theta = \frac{7}{25}$ and θ is in Quadrant I, find $\cos\theta$.

a. $-\frac{7}{25}$

b. $\frac{24}{25}$ ✓

c. $-\frac{24}{25}$

d. $\frac{7}{25}$

$\sin^2\theta + \cos^2\theta = 1 \Rightarrow \cos\theta = \pm\frac{24}{25}$. In Quadrant I, cosine is positive: $\frac{24}{25}$.

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